

Acupotomy for Cervical Spondylosis-Related Neck Pain: A Study Protocol for a Randomized Controlled Trial

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Background: Cervical spondylosis (CS) is a common source of neck pain and imposes a substantial global health burden. Conservative treatments, including oral medication, physical therapy, and rehabilitation exercises, are widely used in clinical practice, but their effects are often limited or slow to become apparent. Acupotomy, a minimally invasive intervention that combines acupuncture principles with soft-tissue release techniques, may alleviate CS-related neck pain by releasing local adhesions and reducing muscle tension. However, methodologically rigorous evidence from randomized controlled trials regarding its efficacy and safety remains limited.

Patients and Methods: This study is a prospective, assessor- and statistician-blinded, randomized parallel-group trial involving patients with CS-related neck pain. The primary hypothesis is that an ultrasound-guided acupotomy-based treatment strategy is more effective than standard medication in reducing pain intensity in patients with CS-related neck pain. Eligible participants will be randomly assigned in a 1:1 ratio to either an acupotomy group or a control group. To strengthen methodological rigor and minimize potential assessment bias, the trial incorporates assessor and statistician blinding, standardized ultrasound guidance during the acupotomy procedure to improve precision and safety, and a predefined sample size calculated based on statistical power analysis. The intervention consists of a 3-week treatment period, followed by a 12-week follow-up period after completion of treatment. The primary outcome will be pain intensity assessed using the Visual Analog Scale (VAS). Secondary outcomes will include the Neck Disability Index (NDI), cervical range of motion (ROM), and health-related quality of life as measured by the 36-item Short Form Health Survey (SF-36). The primary efficacy analysis will be performed using the full analysis set, defined as a modified intention-to-treat population.

Conclusion: This trial is designed to evaluate the comparative effectiveness and safety of an ultrasound-guided acupotomy-based treatment strategy compared with celecoxib for CS-related neck pain. The findings of this study are expected to provide clinically relevant information regarding the feasibility, safety, and potential clinical value of this treatment strategy.

Trial Registration: International Traditional Medicine Clinical Trial Registry (ITMCTR2026000837). Registered on 31 March 2026. The protocol was approved by the Ethics Committee of the First Affiliated Hospital of Shandong First Medical University on 11 March 2026 (YXLL-KY-2026(062)).

Keywords: acupotomy, cervical spondylosis, cervical spondylosis-related neck pain, randomized controlled trial

Introduction

Neck pain is a prevalent musculoskeletal disorder, with cervical spondylosis (CS) representing a major underlying cause.¹⁻⁴ CS-related neck pain is characterized by pain localized in the cervical region, arising from degenerative

changes in the intervertebral discs and associated structural abnormalities, including osteophyte formation, ligament hypertrophy or calcification, and facet joint dysfunction.^{5,6} Although these structural alterations typically develop gradually over years or decades, patients often seek medical attention during acute or subacute exacerbations triggered by mechanical loading, sustained postural stress, or minor trauma. CS-related neck pain imposes a substantial socio-economic and psychological burden.^{7–10}

Conservative management remains the mainstay of treatment for CS-related neck pain.^{11,12} Current therapeutic approaches primarily include pharmacological interventions, particularly non-steroidal anti-inflammatory drugs (NSAIDs), along with physical therapy and structured rehabilitation programs.^{13,14} Although NSAIDs are commonly prescribed for symptom relief, their prolonged use is associated with potential gastrointestinal, cardiovascular, and other adverse effects, which may restrict their long-term clinical application.^{15–19} Consequently, there is an increasing clinical need to explore safer and more effective therapeutic strategies for the management of CS-related neck pain.

Acupotomy is a minimally invasive intervention that combines acupuncture principles with targeted soft-tissue release techniques.^{20–22} Its flat-bladed tip is designed to facilitate both acupoint stimulation and mechanical dissection of adhesions, which may help reduce local tissue tension and alleviate related functional limitations.^{23,24} With the advancement of ultrasound-guided techniques, real-time visualization of anatomical structures may further improve procedural accuracy and safety during acupotomy procedures. Previous clinical and experimental studies have suggested that an ultrasound-guided acupotomy-based treatment strategy may alleviate musculoskeletal pain and improve patients' quality of life.^{25–27} In addition, our preliminary animal studies suggest that acupotomy may indirectly influence intervertebral disc homeostasis by alleviating paraspinal soft-tissue tension and improving local biomechanical loading conditions. These preclinical findings provide a biological rationale for further clinical investigation of acupotomy-based interventions in spinal degenerative disorders. However, this indirect mechanistic linkage requires further validation through dedicated translational studies.^{28,29} Importantly, despite these promising findings, methodologically rigorous evidence from randomized controlled trials evaluating an ultrasound-guided acupotomy-based treatment strategy for CS-related neck pain remains limited, particularly in comparison with commonly used first-line NSAID therapy. Previous studies have been constrained by methodological limitations, including non-randomized designs, small sample sizes, heterogeneous comparator interventions, and inadequate blinding, which may limit the interpretation and generalizability of their findings. Therefore, the present study is designed as a prospective, randomized, assessor- and statistician-blinded parallel-group controlled trial to evaluate the comparative effectiveness and safety of an ultrasound-guided acupotomy-based treatment strategy compared with celecoxib in patients with CS-related neck pain. The primary hypothesis is that this treatment strategy may provide clinically meaningful improvements in pain intensity and functional outcomes compared with NSAID therapy. This trial incorporates several methodological features, including randomized allocation, assessor and statistician blinding, a standardized NSAID comparator, and sample size determination based on statistical power analysis, to enhance the methodological quality and clinical interpretability of the planned evaluation.

Material and Methods

Study Design

This study is designed as a prospective, randomized, assessor- and statistician-blinded, parallel-group clinical trial. A total of 104 participants will be randomly assigned in a 1:1 ratio to either the acupotomy group or the control group. The study protocol was developed in accordance with the SPIRIT guidelines and will be reported following the CONSORT statement.^{30,31} Figure 1 presents a flow chart of the study design.

Participants

Participants will be recruited from The First Affiliated Hospital of Shandong First Medical University and Shandong Provincial Qianfoshan Hospital. Eligible participants will be screened according to predefined inclusion and exclusion criteria and will provide written informed consent prior to enrollment.

Inclusion Criteria

Participants must meet all of the following criteria:

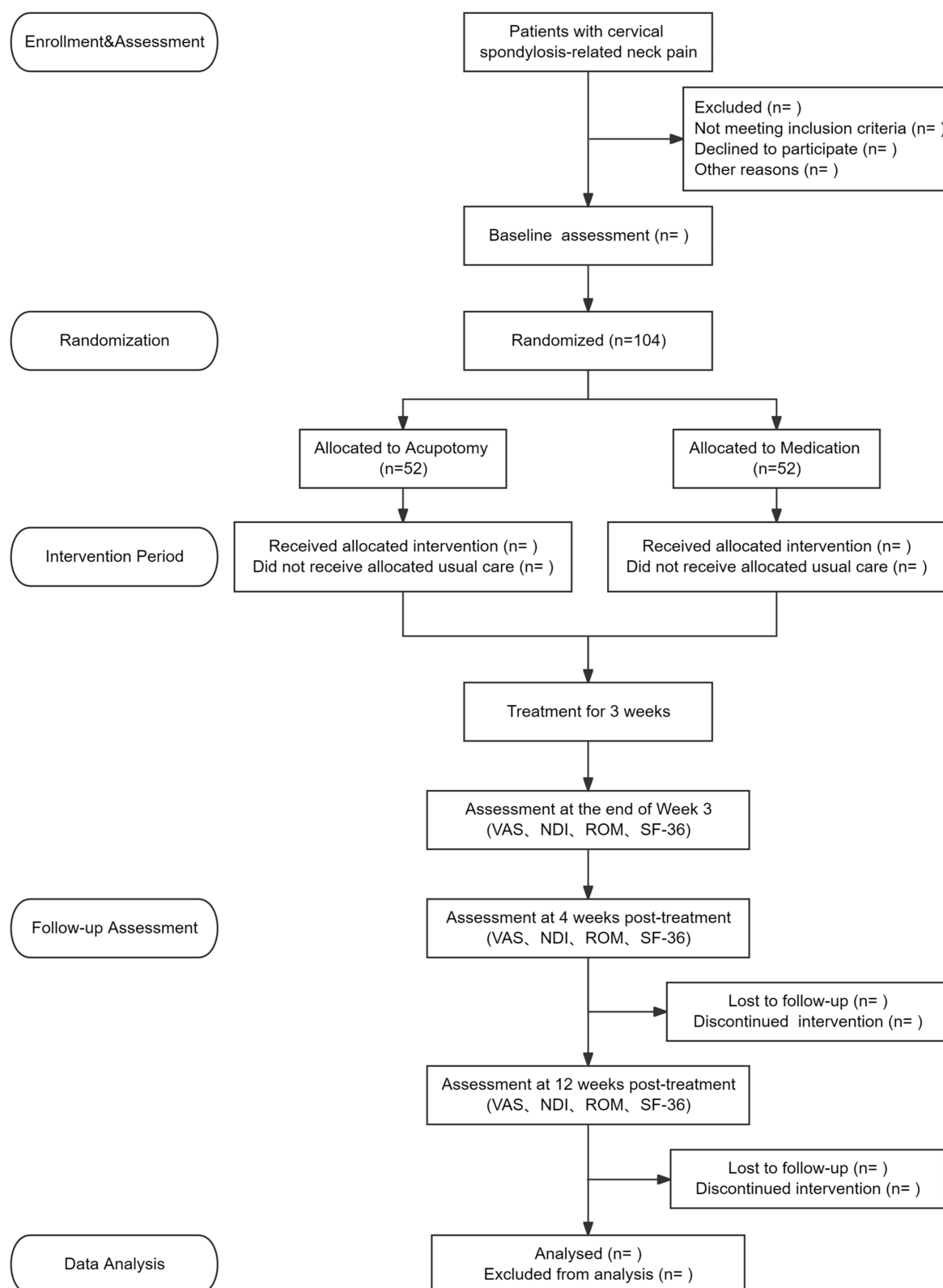


Figure 1 Flow diagram of the trial procedure.

1. Diagnosis of CS-related neck pain confirmed by clinical symptoms and imaging findings.³²
2. Age between 18 and 70 years.
3. VAS score ≥ 4 .
4. Symptom duration between 2 weeks and <3 months.

Exclusion Criteria

Participants will be excluded if they meet any of the following criteria:

1. Presence of neurological deficits or serious cervical pathology.
2. Skin lesions or infections in the cervical region.
3. Severe systemic diseases, coagulation disorders, or uncontrolled psychiatric conditions.
4. Recent use of analgesics or receipt of relevant therapies.
5. Pregnancy or lactation.
6. Known allergy to study-related materials.
7. History of cervical surgery or prior acupotomy treatment.

Intervention

Participants will be randomly assigned in a 1:1 ratio to either the acupotomy group or the control group. The intervention period will consist of 3 weeks, followed by a 12-week follow-up period after completion of treatment.

Acupotomy Group: Ultrasound-Guided Acupotomy-Based Treatment Strategy

Participants assigned to the acupotomy group will receive an ultrasound-guided acupotomy-based treatment strategy. The procedure will be performed by experienced physicians who have undergone standardized training in acupotomy techniques. All interventions will be conducted under real-time ultrasound guidance (Wisonic Navi S, C5–1B MHz convex array probe) to facilitate accurate targeting of the intended tissues and enhance procedural safety.

Treatment points will include bilateral Naokong (GB19), bilateral Quyuan (SI13), along with 3–4 additional nodular points identified between these locations based on clinical assessment. All acupoints will be located in accordance with the World Health Organization (WHO) standard acupuncture point system and modern anatomy (Table 1 and Figure 2).

Participants will be placed in the prone position, allowing full exposure of the posterior neck and upper back. After marking the puncture sites, standard sterile preparation and draping will be performed. Under ultrasound guidance, local infiltration anesthesia will be achieved by injecting 1 mL of 0.5% lidocaine at each site. A Type I No. 4 acupotomy needle (1.0 mm $\times$ 40 mm; Figure 3) will then be inserted through the designated entry point.

The needle will be advanced carefully under continuous ultrasound visualization to avoid major vessels and neural structures until it reaches the target site indicated by the arrow in Figure 4. During needle advancement, participants may experience sensations such as soreness or distension. The operator will then perform 2–3 controlled longitudinal and transverse release movements. Adequate release will be judged by a noticeable reduction in resistance at the needle tip and loosening of the surrounding tissue, after which the needle will be withdrawn.

After the procedure, local compression will be applied for approximately 3 minutes to achieve hemostasis, followed by application of a sterile dressing. The intervention will be performed once weekly for a total of 3 sessions.

Table 1 Locations of Acupoints

Acupoints	Locations
Quyuan (SI13)	Located in the scapular region, in the depression at the medial end of the supraspinous fossa of the scapula, approximately at the junction of the medial border of the scapular spine and the medial border of the scapula.
Naokong (GB19)	Located on the head, lateral to the upper border of the external occipital protuberance, approximately 2.25 cun (about 3.7 cm) lateral to the midline, in the depression between the upper border of the origin of the trapezius muscle and the occipital bone.

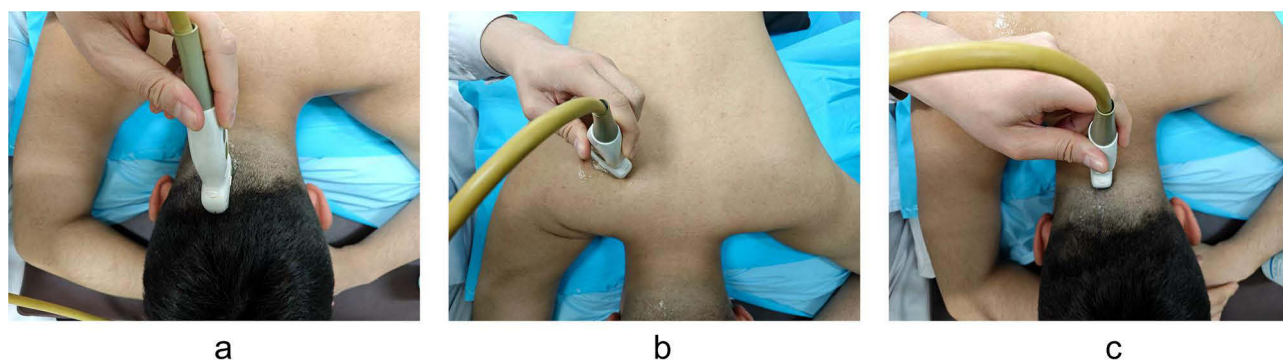


Figure 2 Schematic diagram of operation sites. (a) Location of Naokong acupoint, (b) Location of Quyuan acupoint, (c) Location of nodular soft tissue areas in the neck. The patients provided written informed consent for the publication of their images.



Figure 3 The Type I No. 4 acupotomy (1.0 mm diameter × 40 mm length).

Control Group (Celecoxib Group)

Participants in the control group will receive celecoxib capsules as the active comparator. Celecoxib is a selective cyclooxygenase-2 (COX-2) inhibitor that exerts analgesic and anti-inflammatory effects primarily through inhibition of prostaglandin synthesis. It will be administered at a dose of 200 mg once daily for 3 weeks.

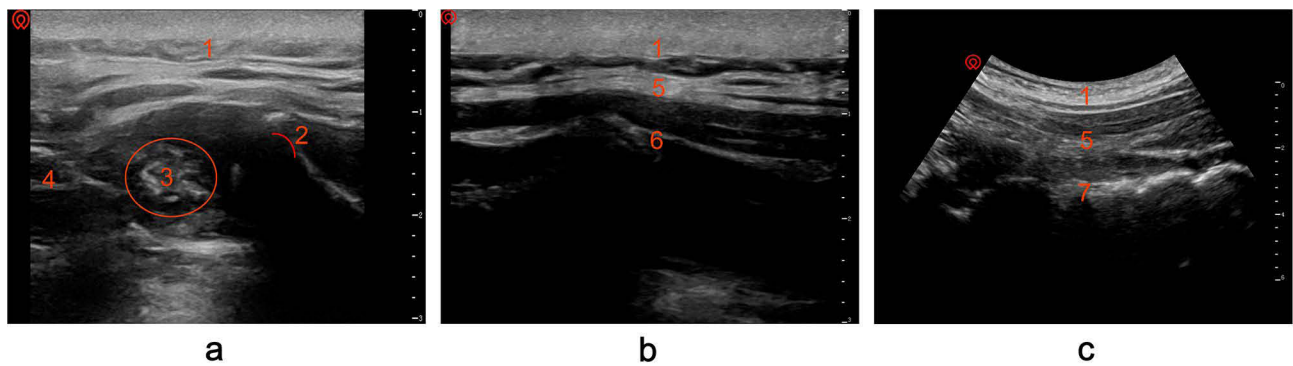


Figure 4 Ultrasound Images of the location for acupotomy release. (a) refers to the ultrasound image of the Qiyuan (SI13) acupoint; (b) refers to the ultrasound image of the Naokong (GB19) acupoint; (c) refers to the ultrasound image of the cervical nodular points.
Notes: red horseshoe like symbol, transducer orientation marker; 1, superficial fascia; 2, superior angle of the scapula; 3, nodular soft tissue areas; 4, levator scapulae muscle; 5, muscle layer; 6, bone surface; 7, facet joint. Among these, positions 3, 6, and 7 are where the acupotomy will reach.

Concomitant Treatments and Rescue Medication

During the intervention period, participants may continue medications for pre-existing medical conditions, provided that these treatments are fully documented and are not intended for the treatment of CS-related neck pain. If participants experience severe intolerable neck pain, oral acetaminophen will be permitted as the only rescue analgesic, with a maximum daily dose of 500 mg and for no more than 3 consecutive days. All use of rescue medication, including dosage, frequency, and indication, will be carefully recorded. The use of additional analgesics, including NSAIDs or other pain-relieving medications specifically for CS-related neck pain, will be prohibited during the intervention period.

During the follow-up period, participants may receive additional treatments for neck pain as clinically required. All post-treatment co-interventions, including pharmacological and non-pharmacological therapies, will be prospectively documented. The type, timing, frequency, and duration of these interventions will be recorded to facilitate the interpretation of medium-term follow-up outcomes.

Study Assessments

Participants will undergo screening and baseline assessments, including eligibility confirmation, informed consent, and baseline outcome measurements, within 1 week before initiation of the intervention. The intervention period will consist of 3 weeks, followed by a 12-week follow-up period after completion of treatment. The complete schedule of assessments is presented in Table 2.

Table 2 Schedule of Enrolment, Intervention, and Assessment

Time Point	Baseline	Treatment Period		Follow-Up Period	
		Before Intervention	The End of Week 3	4 Weeks After Treatment Completion	12 Weeks After Treatment Completion
Enrollment					
Eligibility screen	√				
Informed consent	√				
Allocation	√				
Intervention					
Acupotomy Group					
Medication Group					

(Continued)

Table 2 (Continued).

Time Point	Baseline	Treatment Period		Follow-Up Period	
		Before Intervention	The End of Week 3	4 Weeks After Treatment Completion	12 Weeks After Treatment Completion
Assessments					
VAS		✓	✓	✓	✓
NDI		✓	✓	✓	✓
ROM		✓	✓	✓	✓
SF-36		✓	✓	✓	✓
Adverse events			✓	✓	✓

Notes: The bold text in the table (Enrolment, Intervention, Assessment) are categorized hierarchical headings to delineate the different modules of the research process. “✓” Indicates active intervention procedure for that time point. “_____” indicates the duration of the intervention period from treatment initiation to the end of Week 3.

Abbreviations: VAS, Visual Analog Scale; NDI, Neck Disability Index; ROM, range of motion; SF-36, 36-Item Short Form Health Survey.

Primary Outcome

The primary outcome of this study is the change in neck pain intensity assessed using the Visual Analog Scale (VAS). Measurements will be obtained at baseline, at the end of the 3-week intervention period, and at weeks 4 and 12 after completion of treatment. The primary endpoint is defined as the between-group difference in the change in VAS scores from baseline to the end of the 3-week intervention period.

The VAS is a widely used and validated tool for assessing pain intensity. It consists of a 10-cm horizontal line anchored by “no pain” at one end and “worst imaginable pain” at the other. Participants will indicate their pain level along this line, with higher scores reflecting greater pain intensity.

Secondary Outcomes

In addition to pain intensity, several secondary outcomes will be evaluated, including functional disability, cervical mobility, and quality of life. All secondary outcomes will be assessed at baseline, at the end of the 3-week intervention period, and at weeks 4 and 12 after completion of treatment.

Functional disability will be assessed using the Neck Disability Index (NDI). The NDI evaluates neck pain-related functional impairment through 10 items associated with daily activities, including pain intensity, personal care, lifting, reading, headache, concentration, work, driving, sleep, and recreation. It is widely used to assess functional limitations and treatment-related improvements in patients with CS-related neck pain. Scores are typically expressed as percentages, with higher scores indicating greater disability.

Cervical range of motion (ROM) will be measured using standard clinical tools to provide an objective assessment of movement limitations. Active cervical ROM will be assessed in six directions, including sagittal plane (flexion, extension), coronal plane (lateral flexion), and horizontal plane (rotation), using a goniometer-based measurement approach. Results will be recorded as angular values (degrees) and compared with normal reference ranges (Figure 5). Quality of life will be evaluated using the Short Form-36 Health Survey (SF-36), which is a universal quality of life assessment tool comprising eight dimensions: physical function, physical health, somatic pain, general health, vitality, social functioning, emotional functioning, and mental health. Scores range from 0 to 100, with higher scores indicating better quality of life. This outcome will be used to evaluate the overall impact of the assigned treatment strategies on participants’ physical, psychological, and social functioning.

Safety Evaluation

All adverse events will be carefully recorded throughout the study period. Vital signs will be monitored before and after each intervention session. If any adverse events occur, investigators will assess the participant’s condition, provide appropriate medical management, and document all relevant details, including severity, duration, management measures, and outcomes. Adverse events will be classified according to the Common Terminology Criteria for Adverse Events (CTCAE) version 5.0, and their potential causal relationship with the study interventions will be evaluated.

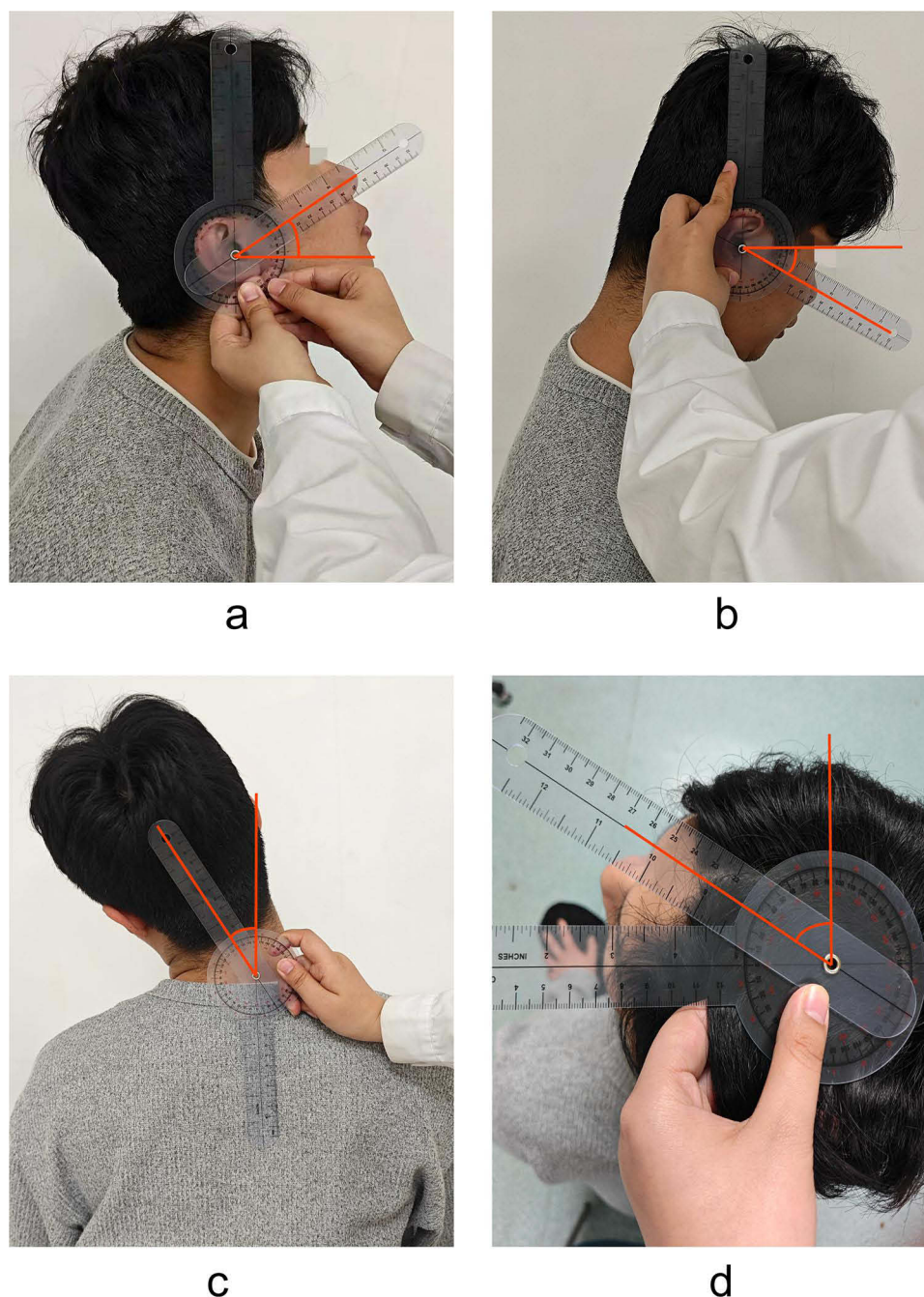


Figure 5 Schematic diagram of cervical spine range of motion. (a) represents the methods for measuring cervical spine extension; (b) represents the methods for measuring cervical spine flexion; (c) represents the methods for measuring cervical spine lateral flexion; (d) represents the methods for measuring cervical spine rotation using a goniometer. The patients provided written informed consent for the publication of their images.

Randomisation Methods

Block randomization will be applied using a fixed block size of 6 to ensure a 1:1 allocation ratio between the acupotomy and control groups. The randomization sequence will be generated by an independent statistician using SPSS software. Allocation concealment will be maintained using sequentially numbered, opaque, sealed envelopes (SNOSE) with tamper-evident labels. These envelopes will be prepared and securely stored by personnel who are not involved in participant recruitment, eligibility assessment, or intervention delivery.

After enrollment, each participant will be assigned a unique randomization number according to the predefined randomization sequence. The participant's screening number and initials will be recorded by the envelope custodian, who will then open the corresponding envelope to reveal the group assignment. The allocation result will subsequently be communicated to the treating investigator. The allocation sequence will remain concealed until interventions are assigned.

Blinding

Due to the nature of the interventions, blinding of participants and treating physicians will not be feasible. Therefore, this trial will adopt an assessor- and statistician-blinded design. Outcome assessors and statisticians will remain blinded to group allocation throughout the study period. Group assignments will be coded as "Group A" and "Group B" in the study database and will remain concealed until completion of the statistical analysis.

Blinding Assessment

The success of blinding will be evaluated after completion of the follow-up period using a questionnaire administered to outcome assessors to determine whether they can correctly guess treatment allocation. After completion of follow-up, database lock, and finalization of the statistical analysis plan, treatment allocation will be unblinded. In the event of a serious adverse event (SAE) or other medical emergency requiring knowledge of treatment allocation, emergency unblinding will be performed through an independent third party in accordance with predefined procedures.

Sample Size Calculation

The sample size was calculated based on the primary outcome measure, namely the between-group difference in change in VAS scores. Because no previous randomized controlled trial has directly compared an ultrasound-guided acupotomy-based treatment strategy with celecoxib for CS-related neck pain, we selected a between-group difference (δ) of 1.5 VAS points as a pragmatic planning assumption. This value was informed by published estimates of the minimal clinically important difference (MCID), as a 1.5-point reduction in VAS score has been widely used and considered clinically meaningful for pain improvement.^{33–35} The estimate was also supported by our own pilot data, rather than simply assuming equivalence between the individual-level MCID and the expected between-group treatment effect. Based on preliminary pilot data, the standard deviation (σ) was estimated as 2.1, derived from the pooled standard deviation of the Week 3 VAS change scores (2.08, conservatively rounded to 2.1). With a two-sided significance level of $\alpha = 0.05$ and a statistical power of 90% ($\beta = 0.10$), the required sample size was calculated using the formula for comparing two independent means: $n = \frac{2(Z_{\alpha} + Z_{\beta})^2 \times \sigma^2}{\delta^2}$. The calculated sample size was 42 participants per group. Considering an anticipated dropout rate of 19%, the total sample size was adjusted as follows: total sample size = $(42 \times 2) / (1 - 0.19) = 104$ participants. Therefore, participants will be randomly assigned in a 1:1 ratio to the acupotomy group and the control group, with 52 participants in each group.

Data Collection and Management

All study data, including efficacy outcomes (VAS, NDI, SF-36, and ROM), safety indicators (adverse events and laboratory test results), and participant information (demographics and medication history), will be recorded using paper-based case report forms (CRFs). The completed CRFs will be regularly reviewed by study personnel to ensure completeness, accuracy, and consistency of the collected data. Any incomplete or non-compliant entries will be promptly identified and addressed to maintain data integrity and traceability. All CRFs will be signed and dated by the responsible investigator. Within 1 week after completion of each participant's follow-up assessment, the completed CRFs and relevant source documents will be submitted to the principal investigator for review and verification.

Data Analysis Plan

All statistical analyses will be performed using SPSS version 26.0. A two-sided P value <0.05 will be considered statistically significant.

The intention-to-treat (ITT) principle will guide the overall analysis strategy, whereby participants will be analyzed according to their originally assigned treatment groups regardless of treatment adherence or protocol deviations. The full analysis set (FAS) will serve as the primary efficacy analysis population and will be defined as all randomized participants who receive at least one study intervention and have at least one post-baseline efficacy assessment. Thus, the FAS represents a modified ITT population that preserves the principle of analyzing participants according to their randomized treatment groups while excluding participants without available post-baseline efficacy data. The per-protocol set (PPS) will include participants who complete the study according to predefined protocol requirements, receive adequate exposure to the assigned intervention, and have no major protocol deviations. PPS analyses will be performed as sensitivity analyses to evaluate the robustness of the primary findings.

Continuous variables will be assessed for normality and presented as mean $\pm$ standard deviation (SD) or median (interquartile range, IQR), as appropriate. The primary endpoint, defined as the between-group difference in the change in VAS score from baseline to Week 3, will be analyzed using analysis of covariance (ANCOVA), with treatment group as a fixed effect and the corresponding baseline value as a covariate. Continuous secondary outcomes will be analyzed using the same ANCOVA model where appropriate. Repeated measurements will be analyzed using generalized estimating equations (GEE) to account for within-participant correlations over time. Categorical variables will be summarized as frequencies and percentages and compared using the chi-square test or Fisher's exact test, with Bonferroni correction applied for multiple comparisons where appropriate.

Missing data for the primary outcome will be handled using multiple imputation (MI) within the FAS under the assumption that data are missing at random. Pooled estimates from the imputed datasets will be reported. Sensitivity analyses, including PPS analyses and analyses using conservative assumptions for missing outcomes, will be performed to evaluate the robustness of the primary analysis results.

Monitoring and Quality Control

Safety data will be periodically reviewed by an independent Data Safety Monitoring Board (DSMB). DSMB members will have no involvement in participant recruitment, intervention delivery, outcome assessment, or statistical analysis to ensure independent oversight. The DSMB will provide recommendations regarding trial continuation, modification, or early termination based on predefined safety and efficacy criteria. The principal investigator will implement the DSMB's decisions.

The trial will be terminated for safety if the incidence of adverse events in the acupotomy group is significantly higher than that in the control group ($P < 0.01$, two-sided) or if the incidence of serious adverse events (SAEs) exceeds 5% in either group. Early termination for efficacy will be considered if the acupotomy group demonstrates statistically significant superiority in the primary outcome (VAS score) compared with the control group. To preserve the overall Type I error rate at the two-sided $\alpha = 0.05$ level, we will apply the Haybittle–Peto stopping boundary, which requires a p -value of < 0.001 for early termination. This extremely conservative threshold effectively eliminates the need for formal α -adjustment, as the chance of crossing this boundary under the null hypothesis is negligible; the final analysis will be conducted at $\alpha = 0.05$ without penalty. Conversely, termination for futility will be considered if no significant difference is observed between groups ($P > 0.05$, two-sided) and conditional power analysis suggests that the probability of achieving the pre-specified effect size ($\delta = 1.5$) with the full planned sample size is less than 20%.

All SAEs will be reported to the Ethics Committee within 24 hours of being recognized by the investigators. The incidence, severity, and potential causal relationship of all adverse events will be recorded and analyzed. Any deviation from the predefined statistical analysis plan will require written justification and prior approval from the principal investigator, statistical analyst, and Ethics Committee.

Discussion

This randomized controlled trial is designed to evaluate the comparative effectiveness and safety of an ultrasound-guided acupotomy-based treatment strategy compared with celecoxib in patients with CS-related neck pain. CS represents an increasingly recognized contributor to neck pain, with its clinical burden potentially amplified by modern lifestyle factors, including prolonged sedentary behavior, sustained forward head posture, and frequent use of electronic devices.^{36–38} The pathophysiology of CS-related neck pain is complex and multifactorial, involving chronic cervical

muscle strain, soft tissue alterations, and altered cervical biomechanics.^{37,39,40} Although degenerative changes typically develop over many years, clinical symptoms may emerge or worsen during episodes of acute or subacute exacerbation, which may be associated with inflammation, mechanical irritation, or soft tissue dysfunction superimposed on pre-existing structural degeneration.⁴¹

Current clinical guidelines recommend NSAIDs, such as celecoxib, as commonly used pharmacological options for the management of CS-related neck pain due to their established analgesic and anti-inflammatory effects.^{42–45} However, while NSAIDs are effective for symptom control, their effects are primarily directed toward pain and inflammatory pathways and may not directly address other potential contributors to persistent symptoms, such as soft tissue dysfunction and altered cervical biomechanics. In addition, concerns regarding adverse effects associated with long-term use and the potential recurrence of symptoms after treatment discontinuation may limit their sustained clinical application.^{19,46} These limitations have prompted interest in complementary minimally invasive approaches that may target additional biomechanical and soft tissue-related factors involved in symptom generation.

It has been hypothesized that acupotomy may exert its effects through mechanical interaction with soft tissues, potentially contributing to the release of adhesions and reduction of myofascial tension.⁴⁷ These biomechanical changes may subsequently contribute to improved local tissue conditions and pain modulation.^{48–50} Ultrasound guidance may further enhance procedural accuracy and safety by enabling real-time visualization of anatomical structures, improving target localization, and potentially reducing procedure-related risks.^{51–53} Collectively, these theoretical and technical considerations provide a rationale for further evaluating the comparative effectiveness and safety of ultrasound-guided acupotomy-based treatment strategies in patients with CS-related neck pain. An ultrasound-guided acupotomy-based treatment strategy may represent a potential minimally invasive approach for managing CS-related neck pain.

However, several limitations of the present study should be acknowledged. First, the absence of a placebo or sham control group limits the ability to fully assess placebo effects. Second, blinding of participants and treating physicians is not feasible because of the nature of the intervention. Although blinded outcome assessment will be implemented to minimize detection bias, the lack of participant blinding remains an inherent limitation, and subjective outcomes, particularly pain intensity scores, should therefore be interpreted with appropriate caution. Third, local infiltration with 0.5% lidocaine will be administered before acupotomy as part of the routine procedural anesthesia to improve participant comfort and facilitate safe delivery of the intervention. Although the analgesic effect of lidocaine is generally transient and may not fully explain potential changes during the 12-week follow-up period, its contribution to early pain relief cannot be completely excluded because no comparable local injection is provided in the control group. Therefore, any potential effects identified in this trial should be interpreted as reflecting the overall effectiveness of the ultrasound-guided acupotomy-based treatment strategy rather than definitive evidence of the isolated mechanical effects of the acupotomy procedure itself. This potential co-intervention bias should be considered when interpreting the findings. Fourth, the single-center design and relatively small sample size may limit the generalizability of the findings. Finally, the 12-week follow-up period may be insufficient to evaluate the long-term efficacy and durability of treatment effects. In addition, outcomes observed at the 12-week follow-up period should be interpreted with caution, as they may reflect both the initial randomized interventions and subsequent clinical management during follow-up. Therefore, week 12 outcomes will be interpreted as reflecting the pragmatic comparative effectiveness of the assigned treatment strategies rather than the isolated effects of the randomized interventions. Future research should address these limitations by adopting a multicenter design, recruiting larger sample sizes, and extending follow-up durations. Where feasible, the inclusion of sham-controlled groups or local anesthetic-controlled procedures would help better distinguish the specific therapeutic effects of acupotomy from nonspecific treatment effects and further clarify its role in the management of CS-related neck pain.

Conclusion

This trial is designed to evaluate the comparative effectiveness and safety of an ultrasound-guided acupotomy-based treatment strategy compared with celecoxib in patients with CS-related neck pain. Although this trial has several limitations, including a single-center design, relatively small sample size, and limited follow-up duration, the findings from this prospective, assessor- and statistician-blinded randomized controlled trial may provide preliminary evidence

regarding the comparative effectiveness and clinical applicability of this treatment strategy. Future large-scale, multi-center studies with longer follow-up periods will be required to further evaluate its clinical effects.

Trial Status

This clinical trial is currently ongoing and actively recruiting participants. The finalized study protocol (version 6.0) received ethical approval from the institutional review board on 11 March 2026 (Approval No. YXLL-KY-2026(062)), and was prospectively registered with the International Traditional Medicine Clinical Trial Registry on 31 March 2026 (Registration No. ITMCTR2026000837). Participant enrollment began on 7 April 2026, and full recruitment is anticipated by October 2026. As of 1 August 2026, a total of 82 participants have been enrolled, and recruitment and follow-up assessments are ongoing.

Abbreviations

CS, Cervical spondylosis; NSAIDs, non-steroidal anti-inflammatory drugs; VAS, Visual Analog Scale; NDI, Neck Disability Index; ROM, range of motion; SF-36, 36-Item Short Form Health Survey; CRFs, case report forms; MI, multiple imputation; FAS, full analysis set; SAEs, Serious adverse events; AEs, adverse events; SNOSE, sequentially numbered, opaque, sealed envelopes; COX-2, cyclooxygenase-2.

Data Sharing Statement

The datasets generated and/or analyzed during the current study will not be publicly available due to patient privacy and ethical considerations but are available from the corresponding author, Liufangming, upon reasonable request.

Ethics Approval and Consent to Participate

The protocol was developed in accordance with the SPIRIT 2025 guidelines, and the results will be reported in accordance with the CONSORT statement. The study is being conducted in the Department of Pain Medicine, The First Affiliated Hospital of Shandong First Medical University and Shandong Provincial Qianfoshan Hospital. The protocol was approved by the Ethics Committee of the First Affiliated Hospital of Shandong First Medical University on 11 March 2026 (Approval No. YXLL-KY-2026(062)), and was prospectively registered with the International Traditional Medicine Clinical Trial Registry on 31 March 2026 (Registration No. ITMCTR2026000837). The study complies with the most recent version of the Declaration of Helsinki. Written informed consent will be obtained from all participants prior to enrollment.

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Author Contributions

All authors made a significant contribution to the work reported, whether that is in the conception, study design, execution, acquisition of data, analysis and interpretation, or in all these areas; took part in drafting, revising or critically reviewing the article; gave final approval of the version to be published; have agreed on the journal to which the article has been submitted Journal of Pain Research; and agree to be accountable for all aspects of the work.

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Disclosure

The authors report no conflicts of interest in this work.

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